Ex.No:16	
DATE:	**NMAP TO DISCOVER LIVE HOSTS USING ARP, ICMP, TCP AND UDP PING SCANS IN TRYHACKME**

1. Aim

To discover live hosts in a network using Nmap by performing ARP scan, ICMP scan, TCP ping scan, and UDP ping scan through a free TryHackMe learning environment.

2. Software Requirements

- Computer or laptop with an internet connection
- Web browser
- Free TryHackMe account
- A free-access lab room with an authorized target
- Nmap terminal in the AttackBox or an authorized lab machine

3. Procedure

Step 1: Create a free TryHackMe account

Open https://tryhackme.com in a web browser. Select Sign Up, register using an available sign-in method, and log in to the account.

Step 2: Select a free room

Visit **https://tryhackme.com/module/nmap** and look for a room or activity related to Nmap and host discovery. Confirm that it is available under the free plan. If the relevant activity requires a subscription, choose another free room that provides a suitable authorized lab target. The learning-path catalogue is available at https://tryhackme.com/paths.

Step 3: Start the lab environment

Open the selected room, read its instructions, and start the target machine. Start the browser-based AttackBox if it is included in the available plan. Note the target IP address shown by the room and open the terminal. If using a local terminal, ensure it can reach the authorized lab target.

Step 4: Verify Nmap installation

Enter the following command in the terminal:

nmap --version

Step 5: Perform ARP scan

Command:

sudo nmap -PR -sn TARGET_IP

Explanation:

- -PR: Uses ARP requests for host discovery.
- -sn: Discovers hosts without performing a port scan.
- TARGET_IP: Replace this placeholder with the IP address of the authorized lab target.

Observation: Record whether Nmap identifies the target as up. ARP discovery generally applies to the same local network segment.

Step 6: Perform ICMP scan

Command:

sudo nmap -PE -sn TARGET_IP

Explanation: -PE sends ICMP Echo Request probes. The -sn option performs host discovery without port scanning.

Observation: Record whether Nmap reports “Host is up”. A firewall may block ICMP requests even when a host is active.

Step 7: Perform TCP ping scan

Command:

sudo nmap -PS80,443 -sn TARGET_IP

Explanation: -PS sends TCP SYN probes to ports 80 and 443. The -sn option disables port scanning.

Observation: Record whether Nmap detects the target as reachable. The result depends on target configuration and firewall rules.

Step 8: Perform UDP ping scan

Command:

sudo nmap -PU53,123 -sn TARGET_IP

Explanation: -PU sends UDP host-discovery probes to ports 53 and 123. The -sn option performs host discovery without port scanning.

Observation: Record the actual output. UDP probes may receive no response even when the target is active.

4. Observation Table

S. No.	Scan Technique	Nmap Option	Observed Result
1	ARP scan	-PR	_____
2	ICMP scan	-PE	_____
3	TCP SYN ping	-PS	_____
4	UDP ping	-PU	_____

Expected observation table

S. No.	**Scan technique**	**Command option**	**Expected observation**
1	ARP scan	-PR	Host detected on the same local network
2	ICMP scan	-PE	Host detected if ICMP response is received
3	TCP SYN ping	-PS	Host detected if a suitable TCP response is received
4	UDP ping	-PU	Host detected if a suitable response is received

5. Result

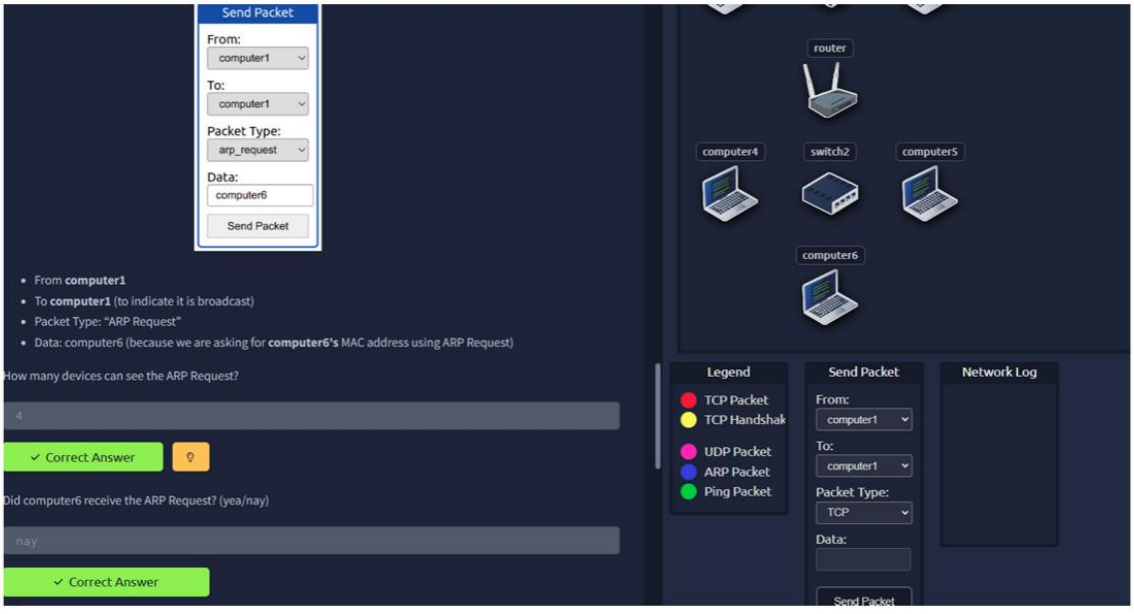
Live host discovery was studied using Nmap in the TryHackMe learning environment. ARP, ICMP, TCP, and UDP probes were used to examine host reachability, and the observations were recorded.

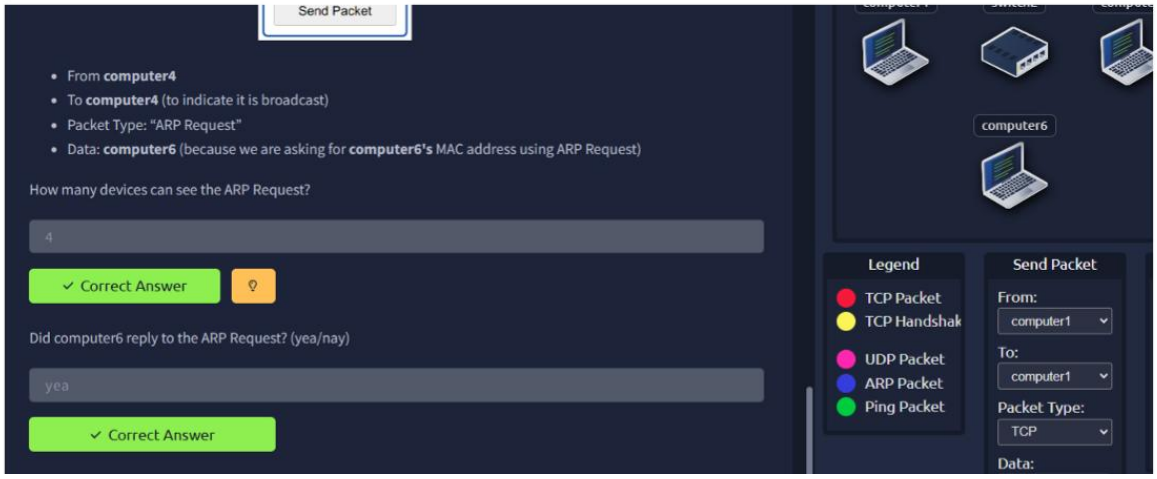
6. Conclusion

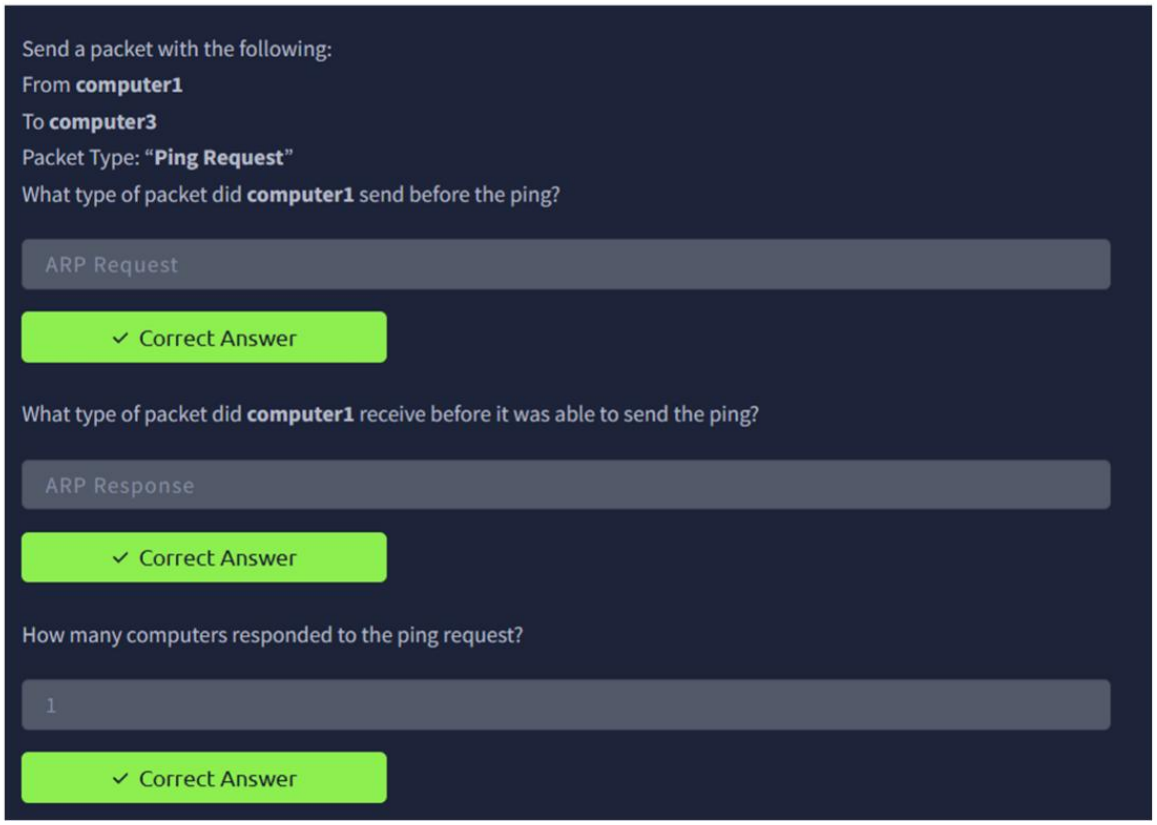
Nmap provides multiple techniques for discovering active hosts. The effectiveness of each technique depends on network topology, target configuration, and firewall rules.

OUTPUT:


Task 1 Introduction
Task 2 Subnetworks
Task 3 Understanding Hosts Discovery through TCP/IP Layer
Task 4 Enumerating Targets
Task 5 Nmap Host Discovery Using ARP
Task 6 Nmap Host Discovery Using ICMP
Task 7 Nmap Host Discovery Using TCP and UDP
Task 8 Using Reverse-DNS Lookup
Task 9 Summary


Send Packet
From:
computer1
To:
computer1
Packet Type:
arp_request
Data:
computer6
Send Packet
From computer1
To computer1 (to indicate it is broadcast)
Packet Type: "ARP Request"
Data: computer6 (because we are asking for computer6's MAC address using ARP Request)
How many devices can see the ARP Request?
4
Correct Answer
Did computer6 receive the ARP Request? (yea/nay)
nay
Correct Answer
router
computer4
switch2
computer5
computer6
Legend
TCP Packet
TCP Handshak
UDP Packet
ARP Packet
Ping Packet
Send Packet
From:
computer1
To:
computer1
Packet Type:
TCP
Data:
Network Log


Send Packet
From computer4
To computer4 (to indicate it is broadcast)
Packet Type: "ARP Request"
Data: computer6 (because we are asking for computer6's MAC address using ARP Request)
How many devices can see the ARP Request?
4
✓ Correct Answer
Did computer6 reply to the ARP Request? (yea/nay)
yea
✓ Correct Answer
computer6
Legend
TCP Packet
TCP Handshak
UDP Packet
ARP Packet
Ping Packet
Send Packet
From:
computer1
To:
computer1
Packet Type:
TCP
Data:


Send a packet with the following:
From computer1
To computer3
Packet Type: "Ping Request"
What type of packet did computer1 send before the ping?
ARP Request
✓ Correct Answer
What type of packet did computer1 receive before it was able to send the ping?
ARP Response
✓ Correct Answer
How many computers responded to the ping request?
1
✓ Correct Answer

Send a packet with the following:

From **computer2**

To **computer5**

Packet Type: "**Ping Request**"

What is the name of the first device that responded to the first ARP Request?

router

✓ Correct Answer

What is the name of the first device that responded to the second ARP Request?

computer5

✓ Correct Answer

Send another Ping Request. Did it require new ARP Requests? (yea/nay)

nay

✓ Correct Answer

Answer the questions below

What is the first IP address Nmap would scan if you provided 10.10.12.13/29 as your target?

10.10.12.8

✓ Correct Answer

How many IP addresses will Nmap scan if you provide the following range: 10.10.0-255.101-125?

6400

✓ Correct Answer

Answer the questions below

How many hosts are found alive after scanning the CONNECTION_IP/24?

1

✓ Correct Answer

Answer the questions below

What is the option required to tell Nmap to use ICMP Timestamp to discover live hosts?

-PP

✓ Correct Answer

What is the option required to tell Nmap to use ICMP Address Mask to discover live hosts?

-PM

✓ Correct Answer

What is the option required to tell Nmap to use ICMP Echo to discover live hosts?

-PE

✓ Correct Answer

Answer the questions below

Which TCP ping scan does not require a privileged account?

TCP SYN Ping

✓ Correct Answer

Which TCP ping scan requires a privileged account?

TCP ACK Ping

✓ Correct Answer

What option do you need to add to Nmap to run a TCP SYN ping scan on the telnet port?

-PS23

✓ Correct Answer

Reverse DNS (rDNS) is the process of resolving an IP address to a hostname (the opposite of normal DNS). Instead of asking "**What is the IP of example.com?**", you ask "**What domain name is assigned to 10.200.6.15?**"

In Nmap, using `-R` forces reverse lookups for all discovered hosts, while `-n` disables them. This helps identify system roles (e.g., mail.company.local, dc01.domain.com) and understand network structure during recon. However, rDNS records aren't always configured or accurate, so results may be missing or misleading. It can also slightly slow scans due to additional DNS queries.

By default, Nmap will look up online hosts; however, you can use the option `-R` to query the DNS server even for offline hosts. If you want to use a specific DNS server, you can add the `--dns-servers DNS_SERVER` option.

Answer the questions below

We want Nmap to issue a reverse DNS lookup for all the possible hosts on a subnet, hoping to get some insights from the names. What option should we add?

-R

